Z Axis Mechanism

Task:

Develop a z axis mechanism for a micro 10 by 10 mm resin 3D printer which will print a Bluesat logo in flight to apogee. The project has an emphasis on print dimensional accuracy and hence the build plate needs to be as stable as possible.

Requirements:

- Should be able to withstand up to 30g forces, spinning induced loads of 6-7 cycles per second and vibrational loads while printing and while the build plate is docked. Additionally it should be able to endure Max Q loads
- Build plate must lock in place when the UV light is hardening filament
- Should be able to deliver a strong enough force to peel the 3D print off the screen's surface. Has to withstand adhesion, gravity and suction forces.

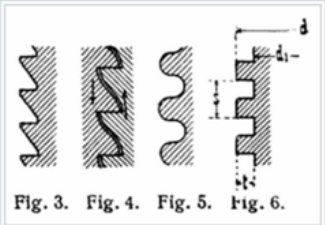
Research:

Lead Screw:

- Translates turning motion into linear motion.
- Large area of sliding contact between male and female members causes large frictional losses
- Used for intermittent low power actuators and commonly used in linear actuators
- Types
- V Thread
- High friction
  - Threads are designed to generate friction to keep the fastener from engaging
  - Like a V timing belt in a car
  - Square thread
    - Most efficient
    - Least friction
    - Used for screws that carry high power
    - Difficult to machine and hence very expensive
  - ACME Thread
    - Favoured in applications needing fast and reliable linear actuators
    - Less efficient than square but easier to manufacture
    - General Purpose and Centralizing varieties feature a thread depth approximately equal to half of the pitch diameter, with centralizing types offering tighter tolerance fits

to minimize wedging under significant radial or side loads.
        - The Stub acme thread has a shallower thread depth—less than half the pitch—ideal for applications where reduced material is advantageous, yet rigidity and high thread engagement remain necessary.
        - 29° thread angle
        - Easier to machine than square but has higher friction
        - Threads are strong, greater load bearing capabilities.
        - Trapezoidal Thread
            - 30 degree thread angle standard in CNC machinery, robotics and industrial automation which require precise linear positioning and repeatability
        - Three types of acme threads are prevalent: General Purpose, Centralizing, and Stub acme threads.
            - General Purpose and Centralizing varieties feature a thread depth approximately equal to half of the pitch diameter, with centralizing types offering tighter tolerance fits to minimize wedging under significant radial or side loads.
            - The Stub acme thread has a shallower thread depth—less than half the pitch—ideal for applications where reduced material is advantageous, yet rigidity and high thread engagement remain necessary.
    - Butress Threads
        - Extremely strong with minimal friction
        - Only work in one direction → not suitable for our application
- For comprehensive calcs. look at the mechanics section of the Wikipedia page

Pros	Cons
• Large load carrying capability • Compactness • Simplicity of design • Ease of manufacture • Large mechanical advantage • Precise and accurate linear motion • Smooth and quiet operation • Low maintenance • Minimal number of parts • Most are self-locking, doesn’t need a braking system (cannot be back-driven)	• 30 - 80% efficiency • High friction and the thread can wear out quickly • Some can't drive backwards



Three types of screw thread used in leadscrews:
3 & 4: buttress thread
5: round thread
6: square thread



An Acme screw

Conclusion: This is the best system for our application. The trapezoidal Acme screw thread is the best choice.

Ball Screw:

- Translates rotational motion into linear motion with minimal friction
- Threaded shaft provides a helical raceway for ball bearings which act as a precision screw
- Can withstand high thrust loads
- 90% efficiency vs 20 - 25% range for Acme lead screw
- Needs a ball recirculation mechanism
  - External ball nut
    - Stamped tube which picks up balls from the raceway by use of a small pick-up finger
  - Internal-button ball nut
    - Employs a machined or cast button-style return which allows balls to exit the raceway track and move one thread then re-enter the raceway
  - Endcap return ball
    - Cap on the end of the ball nut. The cap is machined to pick up balls coming out of the end of the nut and direct them down holes which are bored transversely down the ball nut
  - This mechanism can be plastic as it doesn't experience high loads.
- Read equations section of the Wikipedia for calculations.

Pros	Cons
• 90% efficiency vs 20 - 25% range for Acme lead screw • Eliminate backlash	• Mechanism can be bulky • Significantly more parts and surface interaction • Harder to manufacture

• Minimal friction → extended lifespan, less lubrication req. • Heavier load capacities than lead screws.	• Lower starting torque • **Requires a braking system to prevent backdriving.** • Requires lubrication.



External Ball Nut



Internal-button Ball Nut



Endcap Return Ball Nut

Conclusion: Since it requires a braking system we are adding unnecessary complication to an already complex and small system, we should rule it out.

Rack and Pinion:

- Type of linear actuator that consists of a circular gear (pinion) and a linear gear (rack)
- Rotating the pinion causes the rack to move linearly
- Belt mechanisms work almost the same but the belt can stretch out and break overtime.

Pros	Cons
• High efficiency: Rack and pinion systems have a high mechanical efficiency, typically around 90%, which means that they waste	• Low load capacity: Rack and pinion systems have a relatively low load capacity, meaning that they are not

very little energy in the form of friction or heat. • High speed: Rack and pinion systems can achieve high speeds, making them ideal for applications that require fast linear motion. • Low backlash: Rack and pinion systems have low backlash, which means that there is minimal play or slop between the rack and pinion teeth, resulting in more accurate positioning. • Can withstand harsh environments with high vibrations.	suitable for applications that require high force or heavy loads. • Noise and vibration: Rack and pinion systems can generate noise and vibration during operation, which may be a concern for some applications. • Wear and tear: The teeth on the rack and pinion can wear over time, which can reduce the system's accuracy and efficiency. • **Higher backlash than Lead/Ball screws**

Conclusion: Due to a higher backlash than lead/ball screws this would be a poor choice due to our need for minute precise changes, despite its durability and ability to withstand vibrations.

Voice Coil Actuators:

- Voice coil converts electrical signals directly into linear magnetic force, providing much higher purity of motion compared to DC motors requiring separate drivetrains such as belts or gears
- It consists of a permanent magnetic field assembly (permanent magnets and ferrous steel) and a coil assembly.
- The current flowing through the coil assembly interacts with the permanent magnetic field and generates a force vector perpendicular to the direction of the current.
- The force vector can be reversed by changing the polarity of current flowing through the coil.
- The force generated by the voice coil is proportional to the cross product of the current flowing through the coil and the magnetic flux in the permanent magnetic field, as dictated by Lorentz' force equation.

Pros	Cons
• Simple design and construction, a low hysteresis, being small in size for more efficient design footprint, high accelerations, and a lack of cogging or commutation. • Finer more controlled movement than belts and gears	• Can be expensive • Low forces generated unless large • Needs feedback control.

Type	Cost	Max Stroke	Dimensions	Built in Feedback Sensor
BEI Kimco LA24-20-000A Voice Coil	USD 1188	20 mm class (series typical)	~24 mm diameter class	No
Sensata LA15-16-020A Voice Coil	USD 1800	16 - 20 mm	15 - 30 mm diameter	No
Mini VCM Linear Voice Coil Motor	USD 860	5 to 10 mm		No

Conclusion: Due to Voice Coil's small max stroke length this system will not work for our application.

Thus we will be using a trapezoidal ACME lead screw. Inherently, all lead screws have backlash when they reverse direction. There are a couple of solutions for this.

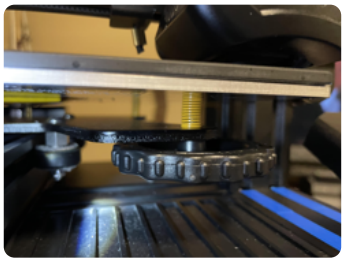
Anti-backlash nuts:

- Two piece nut with a spring compressed between the halves
- Industry standard



Spring Preload:

- Add a spring between the top of the Vat and the build plate, the leadscrew will run inside it.
- You will need to ensure the spring is correctly tensioned.



Dual opposing nuts:

- Lock two standard brass nuts against each other with a slight gap in between.
- Tighten one clockwise and one anticlockwise
- As the nuts wear this will need to be constantly monitored and adjusted
- Don’t know how large vibrations will affect this



Conclusion: Anti-backlash nuts are the simplest way to minimise backlash when changing directions.

Prototyping:

Traditional resin 3D printers have an arm connecting to the lead screw while the other end supports the build plate, as seen below.



This design will not work for our use case as the build plate will be fully enclosed within the vat. Hence the arm will hit the vat wall as it drops down. As such the lead screw needs to directly be connected to the build plate.

Lead Screw Design:

Calculations:

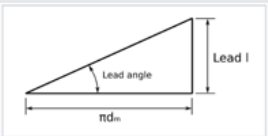


Diagram of an "unwrapped" screw thread

where

- T = torque
- F = load on the screw
- d_m ~~= mean diameter~~ major diameter. $\lambda = \tan\left(\frac{\text{lead}}{\pi d_m}\right)$
- μ = coefficient of friction (common values are found in the adjacent table)
- l = lead
- ϕ = angle of friction $\alpha = \frac{1}{2}$ thread angle.
- λ = lead angle

Considering a conventional trapezoidal lead screw we know that

$$T_{raise} = \frac{F d_m}{2}\left(\frac{l + \pi\mu d_m \sec\alpha}{\pi d_m - \mu l \sec\alpha}\right) = \frac{F d_m}{2}\left(\frac{\mu\sec\alpha + \tan\lambda}{1 - \mu\sec\alpha\tan\lambda}\right)$$

Rearranging this to make l, the lead the subject we get;

$$2T_{raise}(\pi d_m - \mu l \sec\alpha) = F d_m (l + \pi\mu d_m \sec\alpha)$$

$$2T_{raise}\pi d_m - 2T_{raise}\mu l \sec\alpha = F d_m l + F d_m \pi\mu d_m \sec\alpha.$$

$$-2T_{raise}\mu l \sec\alpha - F d_m l = F d_m^2 \pi\mu \sec\alpha - 2T_{raise}\pi d_m$$

$$l(-2T_{raise}\mu\sec\alpha - F d_m) = F d_m^2 \pi\mu\sec\alpha - 2T_{raise}\pi d_m$$

$$l = \frac{F d_m^2 \pi\mu\sec\alpha - 2T_{raise}\pi d_m}{-2T_{raise}\mu\sec\alpha - F d_m}$$

We can make the following assumptions:

$T_{raise} = 0.1$ Nm (a conventional 3D printer needs 0.032 Nm while supporting a 1.25kg gantry. We need to add the effects due to resin.)

Thread angle = 30°
$\therefore \alpha = 15°$.

We know that the peel force can be calculated using Stefan's Viscous Adhesion formula.

$$F_{peel} = 3\pi \times (\text{resin viscosity}) \times (\text{lifting speed}) \times \frac{(\text{radius of cured layer})^4}{2 \times (\text{height between layer and FEP})^3}$$

We can make the following assumptions for this formula.

Height $\approx$ 1/20 mm = 0.05 mm = 0.05×10^{-3} m.
resin viscosity = 200 mPa·s (Elegoo rapid resin at 25°C)
lifting speed $\approx$ 0.10 m/s
$\mu \approx$ 0.14 - 0.16 this is the overlap between steel + steel with oil.
equivalent radius = $\sqrt{\frac{q}{\pi}}$.

Additionally we are only using a single start lead screw and hence pitch = lead.

Lastly we know that the smallest rotation a NEMA 17 Stepper Motor can make is 1.8°. Lets say we want n steps of these to be equivalent to our layer height of 0.05 mm.

0.05 mm : 1.8 n
x : 360°

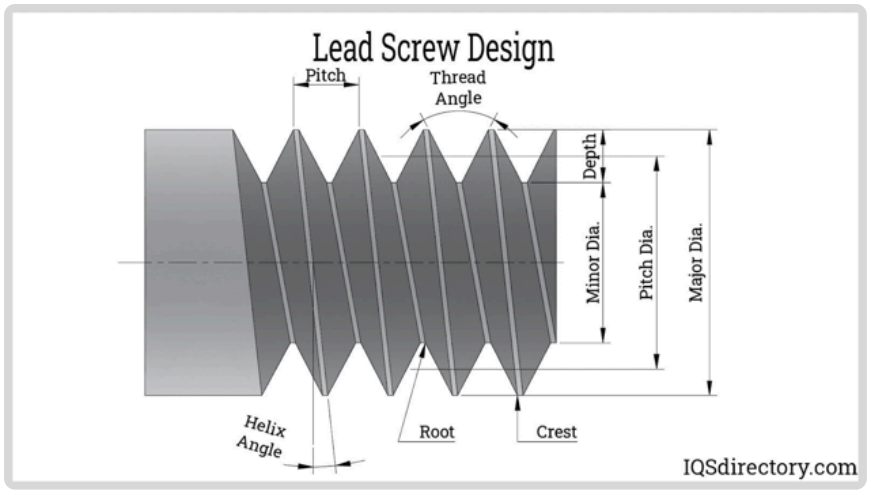
$$x = \frac{360 \times 0.05}{1.8n}$$

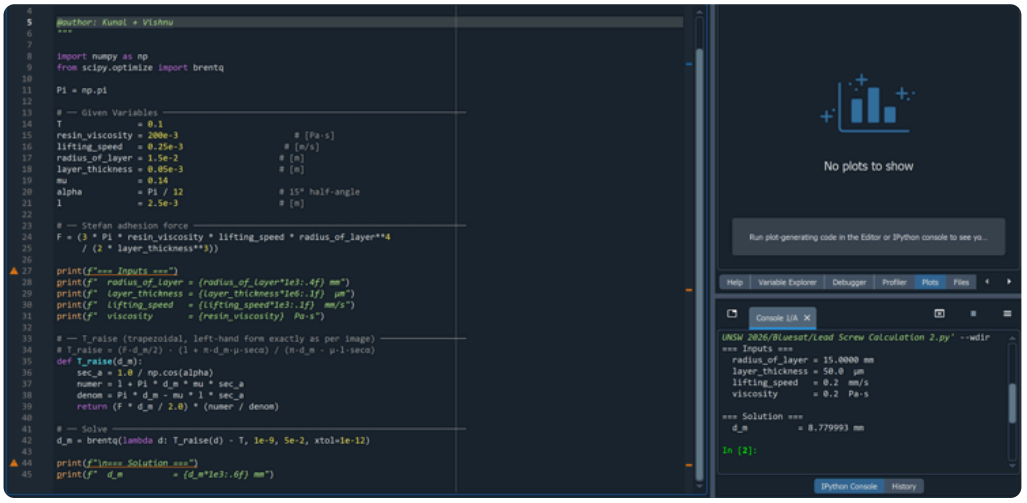
$$x = \frac{10}{n}$$

If we let n = 5 then our pitch and lead is equal to 2 mm.

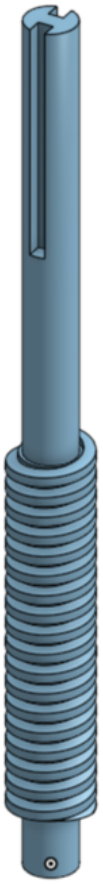
This gives us a major diameter of 8.78 mm.

Since we have made so many untested assumptions we will use the TR 8 x 2 which is the closest to our calculated values as standard trapezoidal lead screws increment by 2 mm.

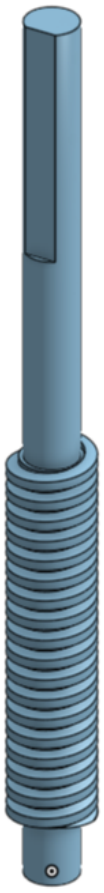

Lead Screw Design
Pitch
Thread Angle
Depth
Minor Dia.
Pitch Dia.
Major Dia.
Helix Angle
Root
Crest
IQSdirectory.com


No plots to show
Run plot-generating code in the Editor or IPython console to see yo...
Help
Variable Explorer
Debugger
Profiler
Plots
Files
Console 1/A
UNSW 2026/Bluesat/Lead Screw Calculation 2.py' --wdir
=== Inputs ===
radius_of_layer = 15.0000 mm
layer_thickness = 50.0 µm
lifting_speed = 0.2 mm/s
viscosity = 0.2 Pa·s
=== Solution ===
d_m = 8.779993 mm
In [2]:
IPython Console
History

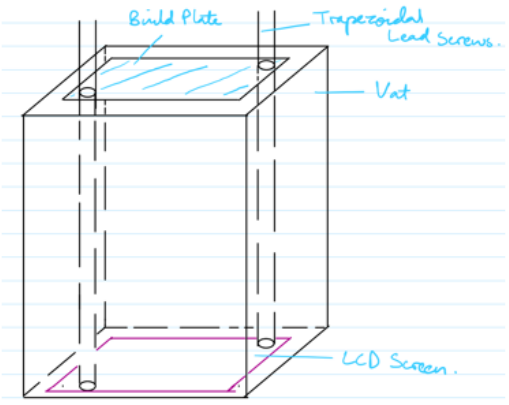
CAD:

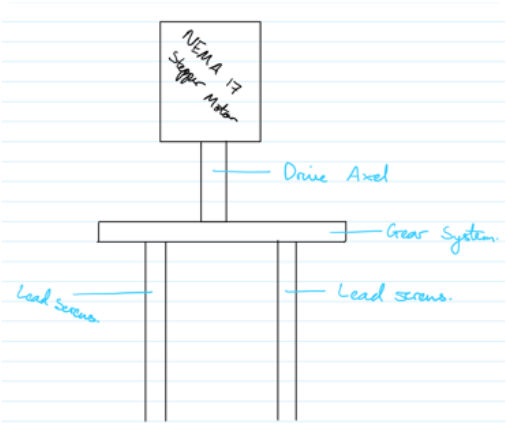


Manufacturing a key like this is very difficult because it requires centering. Instead We can change the key to a D-shaft. This D shaft can be the same as the NEMA Motor.



Prototype 1:

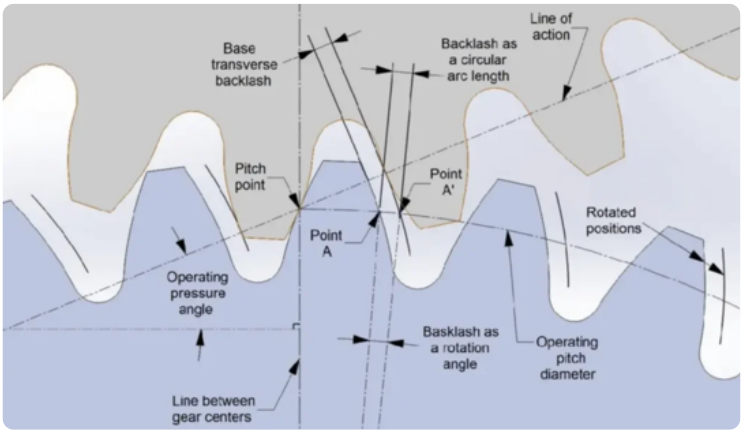




The first image shows how build plate will be secured by two trapezoidal lead screws on opposite corners. These two lead screws will be fixed to the base of the vat or will be slightly suspended. By including two lead screws the build plate has greater support, accounting for the loss of the guide rails and arm, thereby reducing vibration impacts. This design continues to use a single NEMA 17 Stepper Motor as a central drive shaft will connect to the two lead screws via a gear system. This gear system can be attached to the vat lid.

Stepper Motor Connection Prototyping:

Our original prototype suggested the use of gears to connect the drive shaft to the two lead screws. However, gears inherently have backlash which can cause errors in our vertical movement.



To calculate its impact I have done the following calculations of Herringbone Gears which are known for having negligible backlash.

Herringbone Gears are designed to theoretically have no backlash. However, limitations in manufacturing mean that even the most precise gears have an average between 0.01 mm to 0.03 mm.

We know that a linear displacement 2mm will lead to a 2π rad revolution for a TR10x2 Lead screw.

We know that the diameter of our lead screw is 10 mm and hence our radius is 5 mm.

Since $L = r\theta$ we can find the amount the lead screw rotates due to the 0.01 to 0.03 mm backlash.

First we will consider the upper bound.

$$0.03 = 5\theta$$
$$\theta = \frac{0.03}{5}$$

$$2 \text{ mm} : 2\pi$$
$$x : \frac{0.03}{5}$$

$$x = \frac{\frac{0.03}{5} \times 2}{2\pi}$$

$$x = \frac{2 \times 0.03}{5 \times 2\pi}$$

$$x = \frac{0.03}{5\pi}$$

We can know calculate the percentage change in layer height given we know our layer height will be 0.05 mm.

$$1 - \frac{\left| 0.05 - \frac{0.03}{5\pi} \right|}{0.05} = 3.8197\%$$

Hence the upperbound of average backlash in a Herringbone gear is causing a a vertical displacement 38% of our layer height.

Doing the same calculation for the lower bound we get

$0.01 = 5\theta$

$\theta = \frac{0.01}{5}$

$2\text{ mm} : 2\pi$

$x : \frac{0.01}{5}$

$$x = \frac{\frac{0.01}{5} \times 2}{2\pi}$$

$$x = \frac{0.01 \times 2}{5 \times 2\pi}$$

$$x = \frac{0.01}{5\pi}$$

$$\%\text{ change} = 1 - \frac{\left|0.05 - \frac{0.01}{5\pi}\right|}{0.05} = 1.2732\ldots\%$$

∴ We know that implementing a Herringbone Gear System will lead to a vertical displacement of

$1.273\% - 3.820\%$

as a percentage of our layer height.

The other option is using a Pulley System. Pulley systems can generate backlash like effects because of

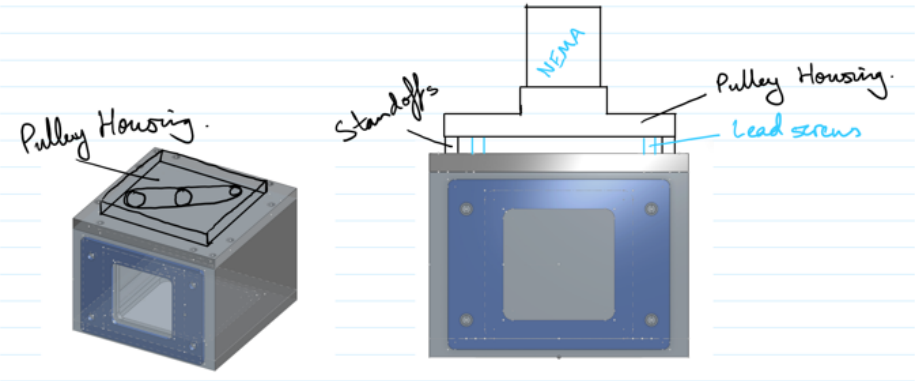
- The belt being poorly tensioned can cause it to slip.
- The belt stretching under heavy loads, causing it to lag
- Or the pulley's not being fastened tightly.

All of these can be mitigated with a good design. Hence we will be going with a Pulley system.

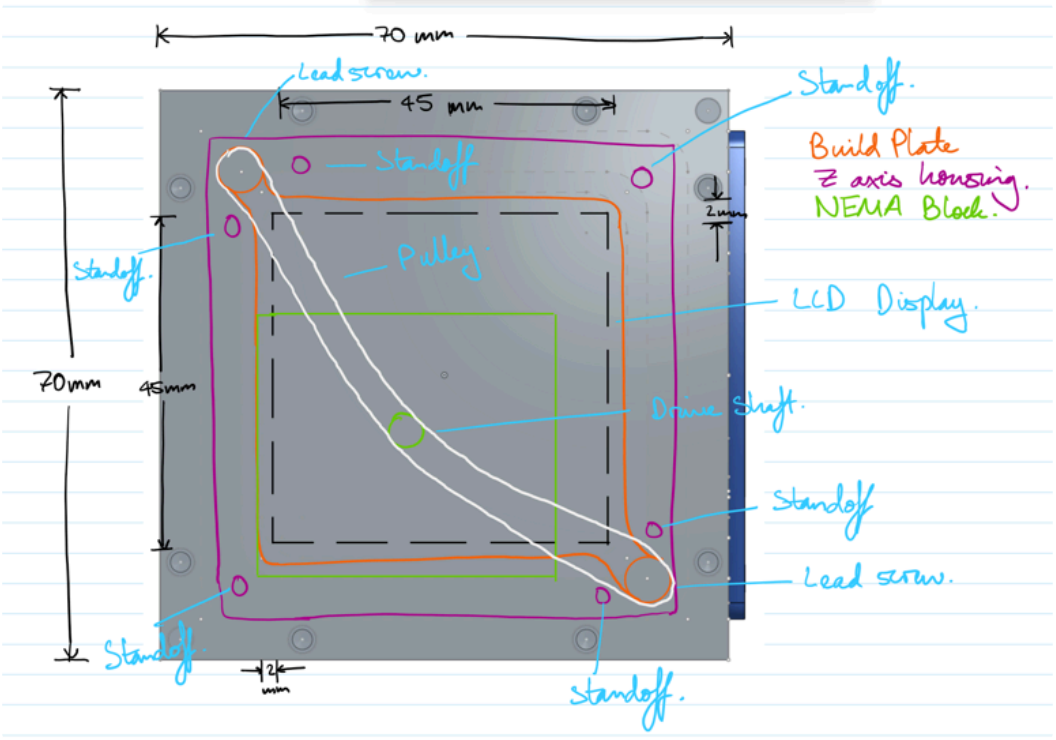
Housing/Fastening Prototyping:

We need to make our 3D printer as easy to use as possible and hence need to make access to model as easy as possible. Hence I am suggesting a modular design in which every system is housed separately but connected together. This way undoing only the bolts for the Vat Lid should be enough to access the 3D print.

I am proposing the following design.

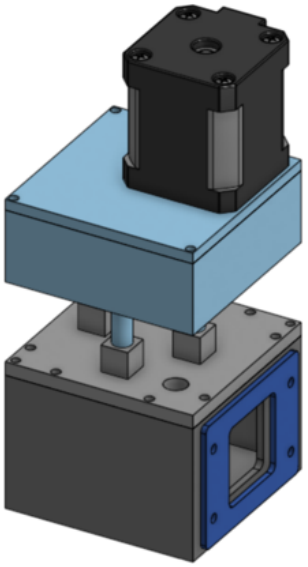


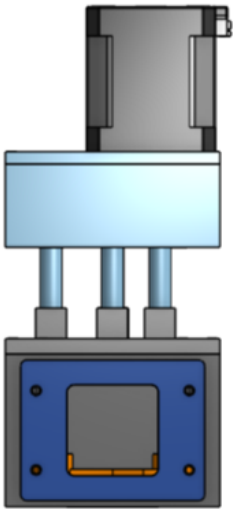
The design consists of central housing block which contains the whole pulley system. The NEMA motor will be connected to the top of this housing and the two lead screws will come out the bottom and into the Vat. The housing will be placed on four standoffs which will be bolted onto the Vat Lid. When we need to fill the Vat or take out our 3D print model we only need to take off the Vat Lid and the whole Z axis mechanism should come off with it. This way we maximise efficiency while keeping the design simple. The standoffs are symmetrical about the diagonal so that the M3 bolt heads do not hit the timing belt.

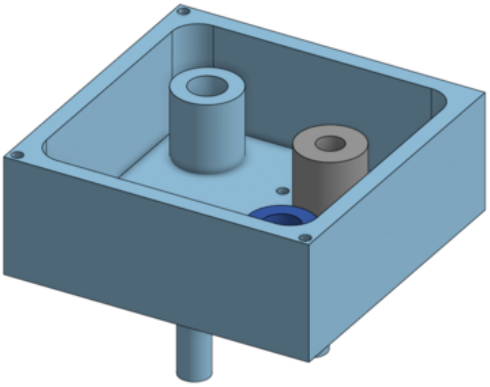


This is a more complex design showing the positioning of all the key elements on the Vat Lid. The pulley may need a tensioner because the pulley's are almost colinear.

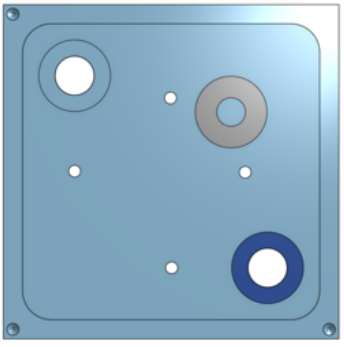
CAD:







The cylinders are representative of the Pulleys



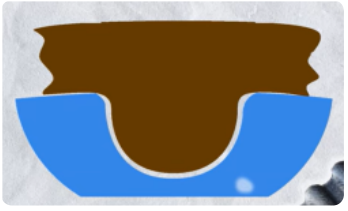
Pulley System Design:

Timing Belt Systems transfer motion and power from one Pulley to another without any slip. There are three types of timing belts.

- Trapezoidal Tooth
  - Precise timing and indexing
  - Minimum backlash
  - Ideal for precision applications



- Circular Teeth
  - High Torque transfer and high load carry capacity
  - Not accurate for precise timing



- Modified Curvilinear
  - High torque capacity
  - Good indexing precision better than High Torque Drive Belts (trapezoidal)



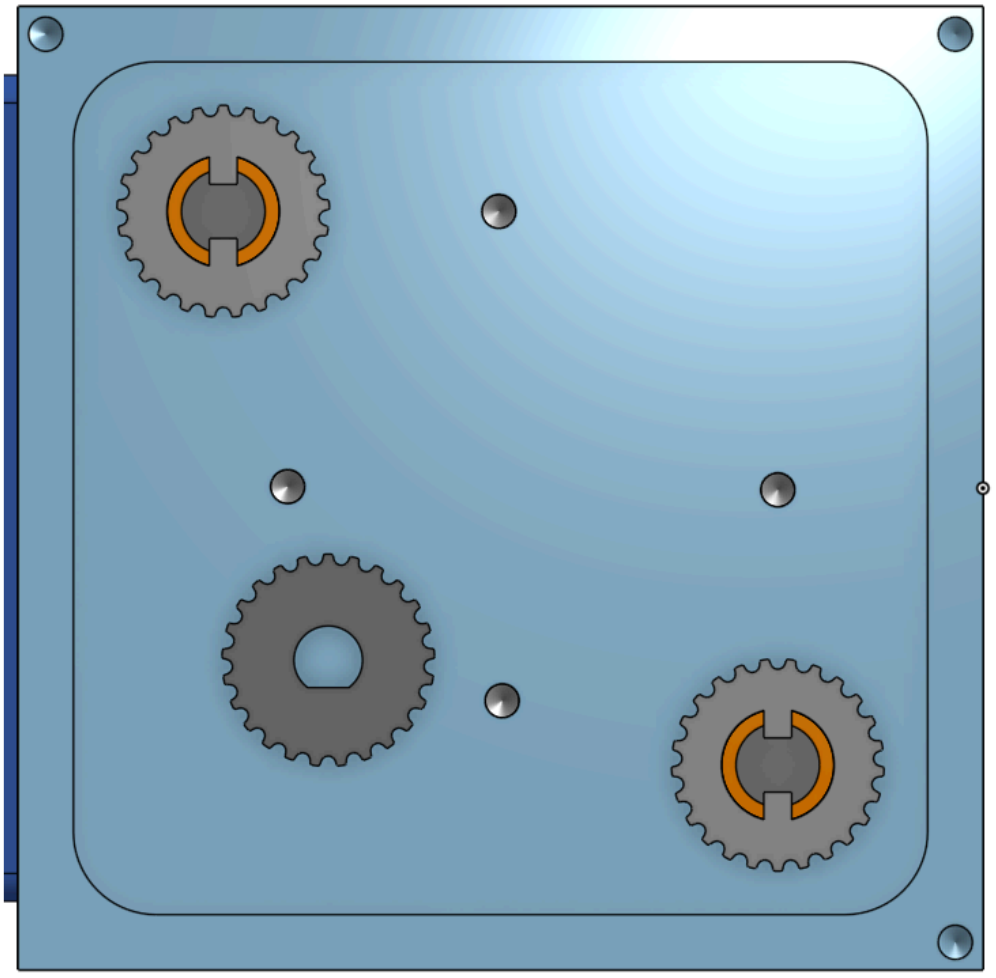
This is a ChatGPT summary for our specific use case. Which highlights Modified Curvilinear timing belts as the best option. This is consistent with the research provided above.

Tooth profile type	Trapezoidal	Curvilinear	Modified curvilinear / circular
Typical pitch	2.032 mm	3 mm	2 mm
Approx belt cost (AU)	~$3–10	~$2–25	~$4–15
Backlash	Low	Moderate	Very low
Positioning accuracy	Good	Moderate	Excellent
Noise	Higher	Lower	Low
Torque capacity	Moderate	High	High
Wear resistance	Moderate	Good	Excellent
Common in 3D printers?	Older printers	Rare	Extremely common
Pulley availability	Very common	Common	Extremely common
Best use case	Cheap/simple motion	High torque drives	Precision positioning
Recommended for resin printer?	⚠️ possible	❌ not ideal	✅ strongly recommended

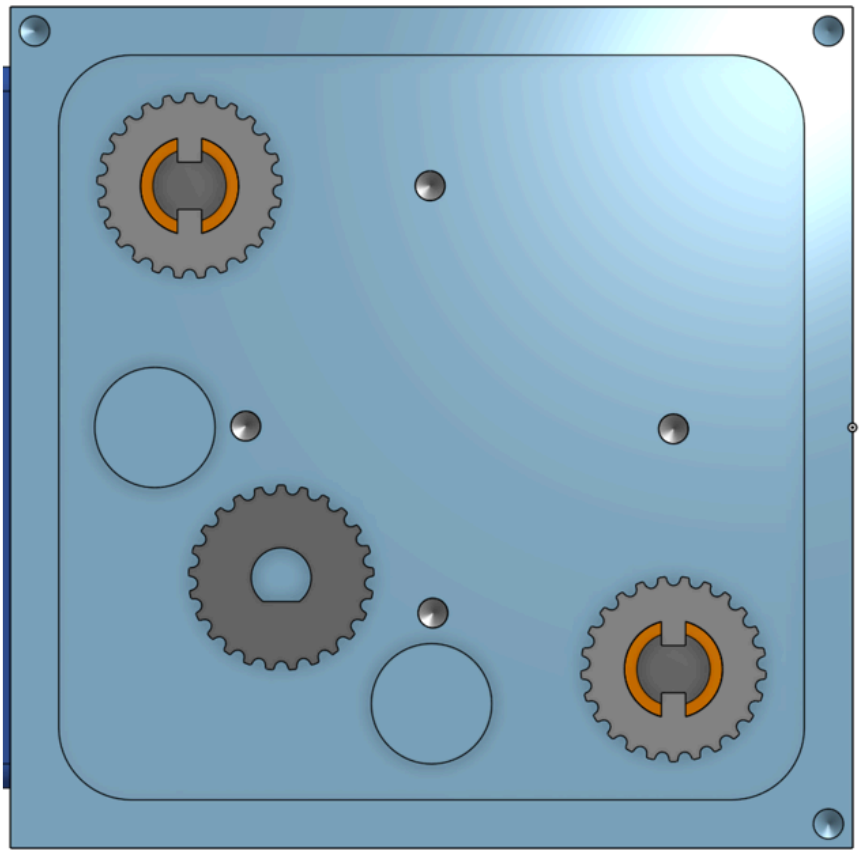
The Gates 2mm Tooth Profile (GT2) is industry standard for 3D printers because it provides smooth operation, precise positioning and low backlash while being able to withstand higher torque than their older trapezoidal counterparts. As such we will use a GT2 timing belt for our application as well.

We now need to consider the speed ratios between the drive shaft and the lead screw pulleys as this is directly proportional to size difference between the pulleys.

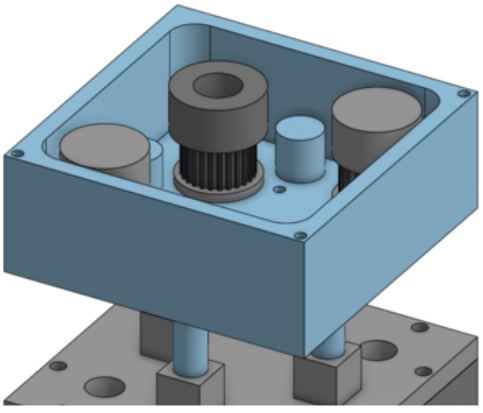
To keep the design simple we should use Pulley's of the same diameter for the driver and driven pulleys. This also makes calculations to determine how much the motor needs to turn to create a vertical displacement of one layer height. However, we must ensure that the wrap angle of the belt around the pulley's is greater than 90° for high torque loads and must be greater than 60° to ensure the timing belts transfer power effectively. We may need to artificially increase this by adding in a set of tensioners.



These are 25 toothed GT2 Pulleys with inbuilt notches to lock into the lead screw and the NEMA Shaft to ensure no slippage. It is clear that there needs to be two tensioners to increase the wrap angle on the drive pulley as it is less than 90° by visual inspection.



I have added a lip to all three pulley's with a 1 mm tolerance to the top of the lid so that the pulley's don't experience vertical displacements which will break the Z axis mechanisms. Furthermore, I have added a set of 2mm spacers on the bottom of each pulley which will be 3D printed as a separate part.

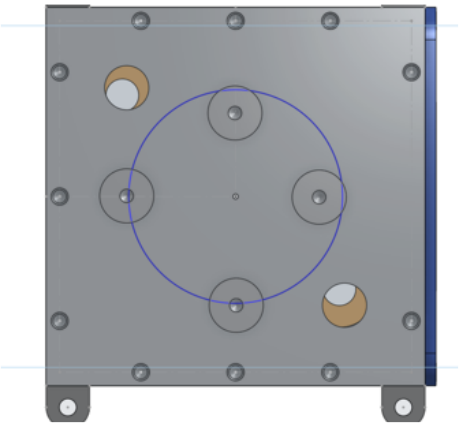


Feedback:

- How the lead screws interface with the Pulley was not clear in the documentation.
- The tensioners need to be redesigned so that they can be adjusted to change the belt tensions easily.



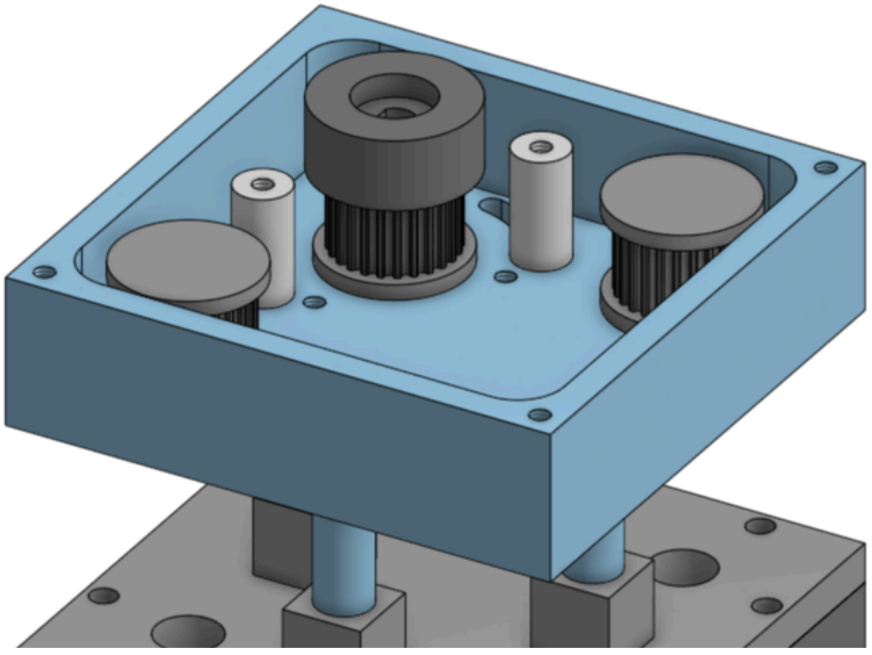
- The drive shaft of the motor needs to have a larger surface area of contact with the drive pulley. This needs to maximised to minimise failure due to bending moments.
- You need to explain why the standoffs are not designed to be symmetrical about the origin but rather the diagonal.

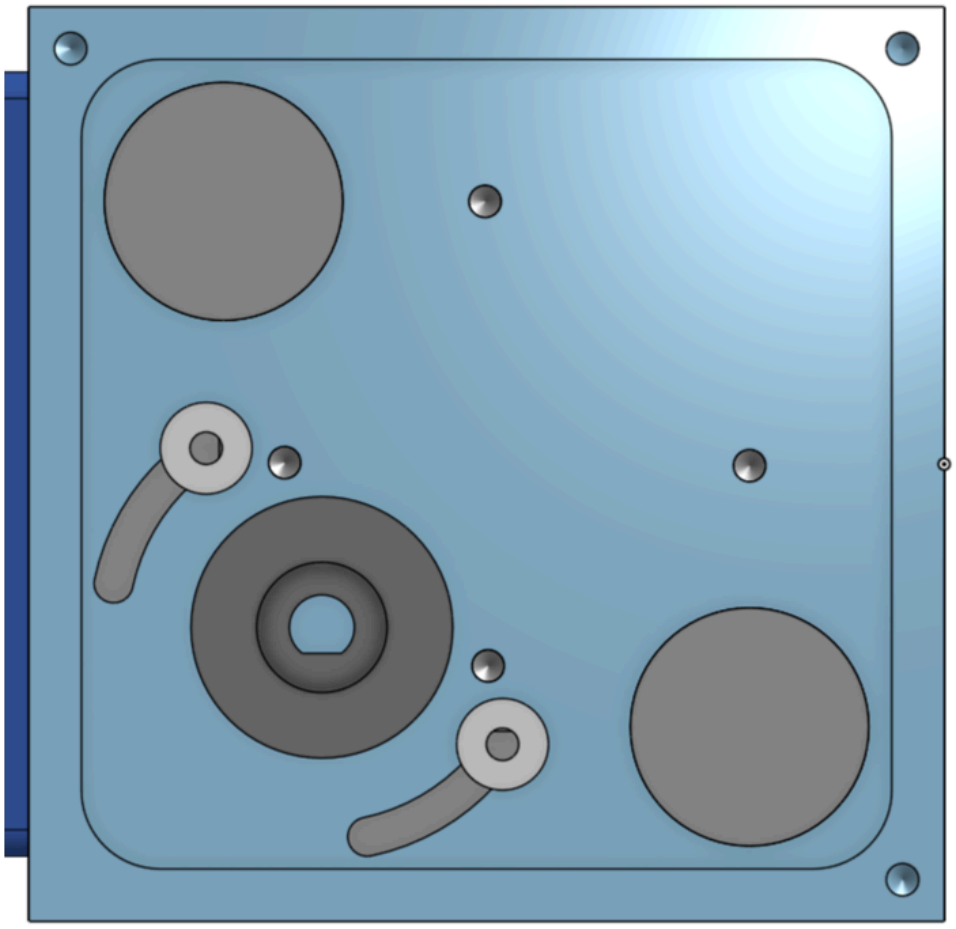


- Redesign the Housing Shape to minimise material usage.
- Rename the parts studios in Onshape.

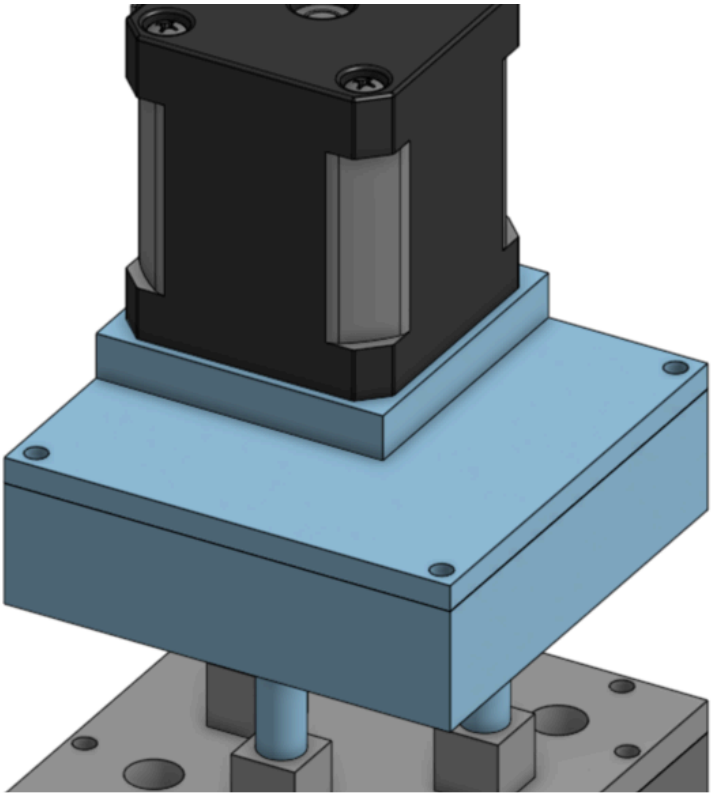
Prototype 2:

Prototype 2 synthesises the feedback with prototype 1 to create a more robust design that satisfies the requirements better.





The tensioner system has been redesigned such that it can slide along the rails and be fixed in with a screw and washer through the bottom, further including a free spinning collar to minimise friction between the belt and the tensioner.



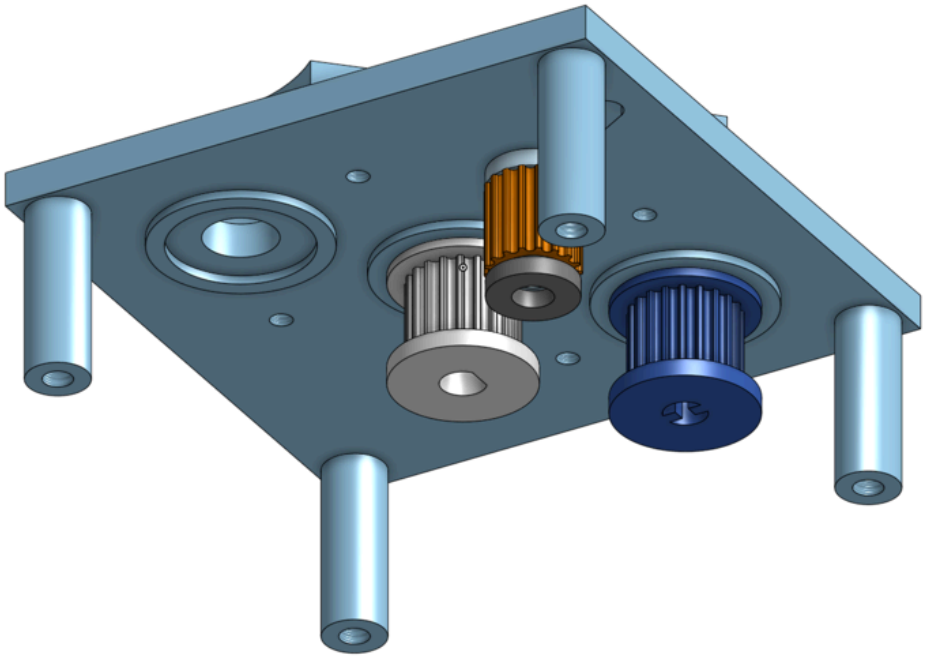
The lid has also been redesigned to have a motor mount which minimises the material usage.

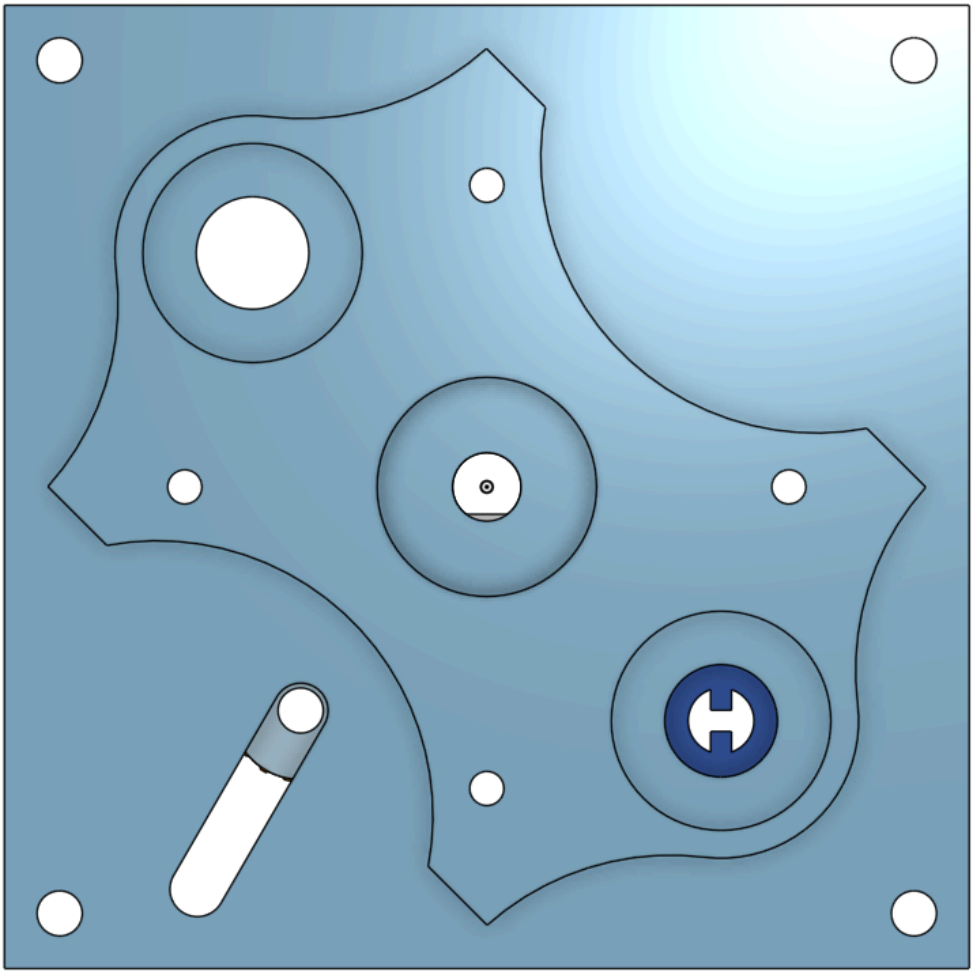
Lastly, the grooves in all the pulley's have been updated so that the lead screws and NEMA drive shaft can be press-fitted into the Pulleys.

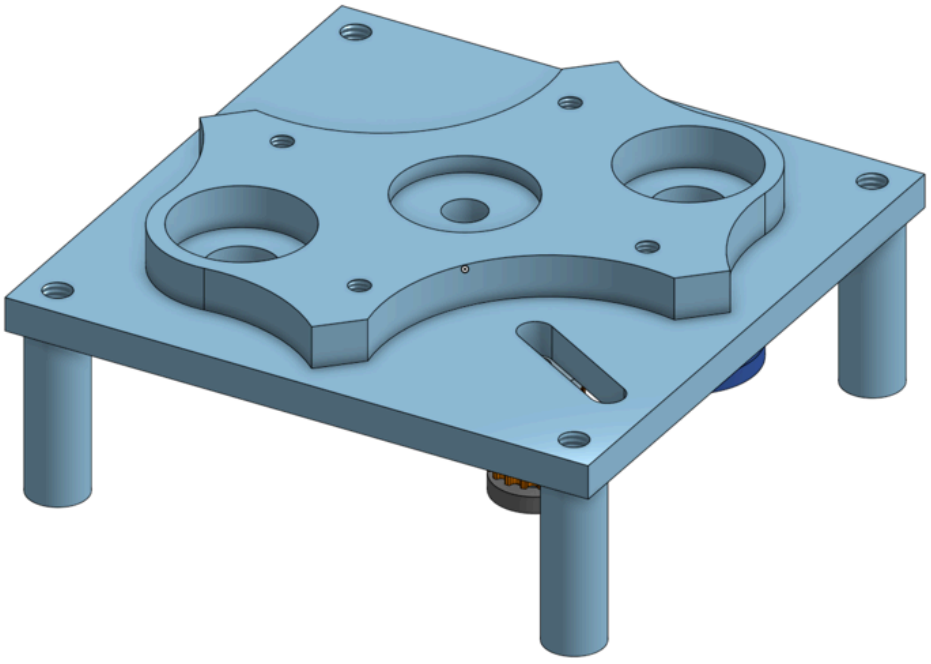
Prototype 3:

To optimize manufacturing time and material usage we are moving to a single plate design in which the motor will be mounted on top with the Pulley's below. This design will utilise deep groove ball bearings to interface between the plate and the lead screw and the base of the Vat and the lead screw. As such any effects of vibration will be minimised and will make it easier to put the lead screw in vertically.

Since we are using TR8X2 lead screws the root diameter will be around 5.5 mm. We will use a lathe to remove the thread and turn down a journal of 5 mm. We will then use a 625 series deep grooved ball bearing.

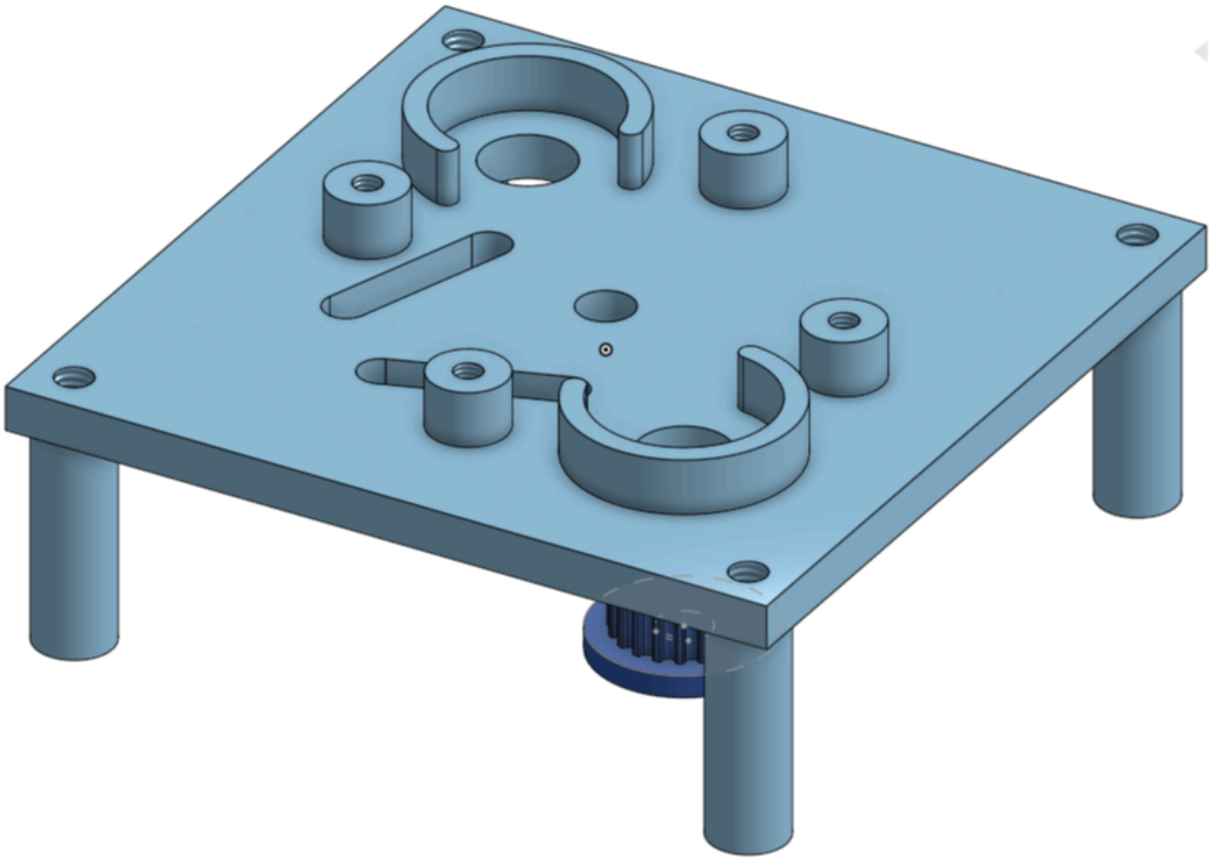


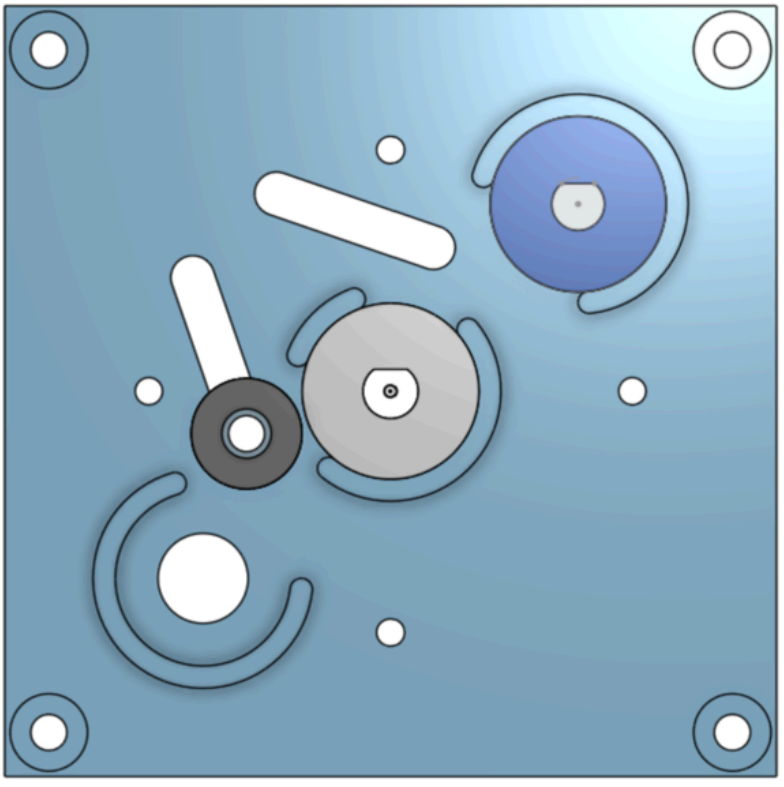




Prototype 4:

Due to double sided timing belts being non existent for GT2 systems. I redesigned to have only a single sided timing belt. This design, while adding another tensioner system, has a higher angle of contact between the belt and the Pulley, thus giving a higher torque output.





Manufacturing:

Lead Screw + Brass Nuts:

Since our calculation had so many untested assumptions we have decided to use the TR8x2 lead screw because it is the closest to our calculated values. We will be using Stainless Steel 304 or 316 for the lead screw material because it is chemically inert. Additionally we will need to use lead screw nuts to connect the build plate to the lead screw. These are usually made out of brass so that they move slowly along the stainless steel lead screw. Brass is not chemically inert and will begin to corrode in the resin. Since the 3D printer will not be used often we can use brass nuts or we can manufacture our own lead screw nuts out of stainless steel. We will need to do a lot of testing to design the lead screw nuts ourselves and we will need to see if the resin will provide sufficient lubrication between the lead screw and the nut.

We can buy a set of two stainless steel 304 lead screws and with two brass nuts provided from Amazon for $34.

2pcs 400mm Tr8X2 Lead Screw with T8 Brass Nut for 3D Printer Machine Z Axis(Acme Thread, 2mm Pitch, 1 Start, 2mm Lead)

We require an extension of the lead screw at the top and bottom to interface with the Pulleys and the bearings. Anyone with Lathe training will need to turn down a journal, i.e. strip off the thread down to 5 mm. From here we can just use light press fitting to interface the lead screw with the driven pulley.

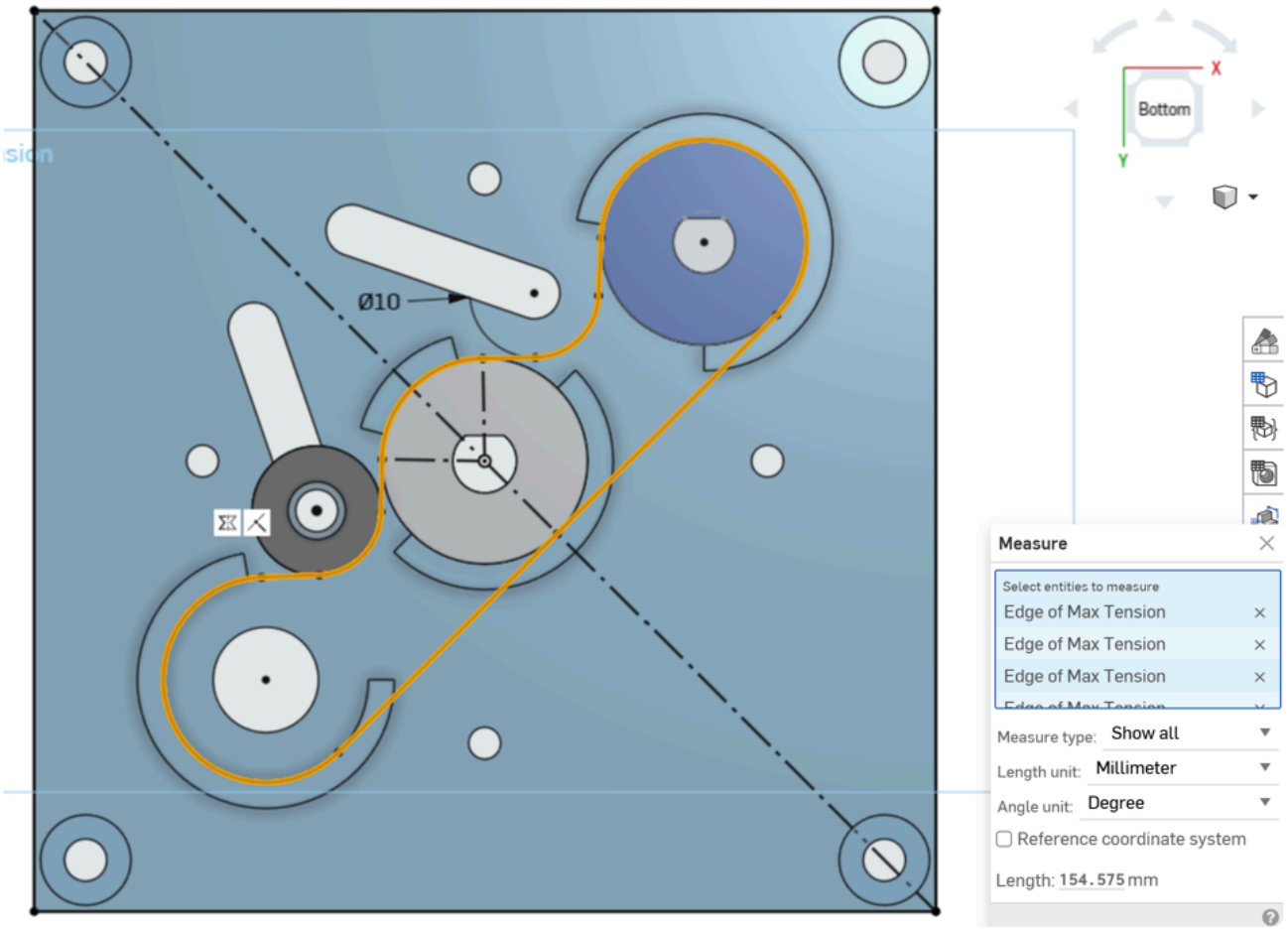
Pulleys/Washers:

The pulleys and washers will be 3D printed using PLA.

Housing:

The Pulley Housing will be machined using a CNC Mill.

GT2 Timing Belt:



GT2 (2mm) Timing Belt (per meter) - Maker Store PTY LTD

Assembly:

Inventory List:

- 4x M4 35 mm Socket Screws to connect the Pulley Housing Frame to the Vat Lid.
- 4x M3 12 mm Socket Screws for NEMA Motor
- 1x M4 16 mm Socket Screws for the Tensioner
- 4x 625 series Bearings
- 1x Build Plate
- 2x Lead Screw Nuts
- 1x Timing Belt (155 mm)
- 2x Tensioner Bases
- 2x Rollers
- 2x M4 20 mm Socket Screws
- 2x M3 Nuts

Steps:

1. Attach NEMA Motor to frame.
2. Light press fit the Drive Pulley onto the NEMA Motor Shaft

3. Light Press Fit the Driven Pulleys onto the lead screws
4. Place the 625 bearings in the top grooves and secure the outer race.
5. Slide lead screws in and fasten to the inner race of the bearings.
6. Attach the tensioner
7. Attach the frame to the Vat Lid
8. Attach the build plate to the lead screw nuts
9. Thread the lead screw nuts onto the lead screws.
10. Fasten the Vat Lid.
11. Start Printing!!!